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Exercise: Day 3

In this problem, you will work through the transition state theory for a bistable harmonic system, such as described in the notes. The reaction coordinate for this system is q . Assume the system has two harmonic wells, described locally by the following potentials,

$$w(q) = \frac{1}{2}m\omega^2(q - q_A)^2 \quad \text{and} \quad w(q) = \frac{1}{2}m\omega^2(q - q_B)^2 + \Delta w$$

And a barrier that is described locally by,

$$w(q) = -\frac{1}{2}m\omega^2(q - q^*)^2 + \delta w^\ddagger$$

1. Evaluate the numerator of k_{AB}^{TST} by computing:

$$\langle \dot{q} \theta(\dot{q}) \delta(q - q^*) \rangle = \frac{1}{Z} \int dq \int d\dot{q} \dot{q} \theta(\dot{q}) \delta(q - q^*) e^{-\beta \left[\frac{1}{2} m \dot{q}^2 + w(q) \right]}$$

Perform the Gaussian integral over velocities. Hint: $\int_0^\infty dv \, v e^{-\frac{1}{2}\beta m v^2} = \frac{1}{\beta m}$

2. Evaluate the denominator $\langle h_A \rangle$ by performing the Gaussian integral over q near q_A . To make solving this problem easier, you should assume that $\beta \delta w^\ddagger \gg 1$.

3. Combine your results from parts 1 and 2 to show:

$$k_{AB}^{\text{TST}} = \frac{\omega}{2\pi} e^{-\beta \delta w^\ddagger}$$

4. Identify the Arrhenius structure in this expression.
 - a. What quantity plays the role of the activation energy?
 - b. What is the physical interpretation of the pre-exponential factor $\omega/2\pi$?
 - c. How does k_{AB}^{TST} depend on temperature in the low- T and high- T limits?

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5. Write down the reverse rate k_{BA}^{TST} (i.e., going from B back to A).

6. Show that the ratio of forward and reverse TST rates satisfies detailed balance:

$$\frac{k_{AB}^{\text{TST}}}{k_{BA}^{\text{TST}}} = e^{-\beta(w(q_A) - w(q_B))}$$

Interpret the right-hand side as an equilibrium constant. What does this tell you about the equilibrium state of a system governed by TST?

7. The true rates satisfy, $k_{AB} = \kappa_{AB} k_{AB}^{\text{TST}}$ and, $k_{BA} = \kappa_{BA} k_{BA}^{\text{TST}}$. By demanding that the true rates also satisfy detailed balance, show that:

$$\kappa_{AB} = \kappa_{BA}$$

Explain this result physically: why must the fraction of recrossings be the same in both directions?

8. Describe two physical scenarios:

- One in which you expect $\kappa \approx 1$ (TST is a good approximation).
- One in which you expect $\kappa \ll 1$ (many recrossing)

What properties of the potential $w(q)$ or the system dynamics would lead to each case?